

POWER GENERATION SOLUTIONS

Technical appendix

Sixteen zone cable packages and the full catalog of published references, with specification numbers and manufacturer documents. Companion to the Accelerating Time to Power e-brochure.

Stock numbers do not indicate inventory. Confirm voltage, insulation level, conductor size, installation rating, reel length, listing, pricing and lead time against the project design.

Applications 01–04. From equipment to scope.

01 Switchyard / GSU / grid tie

Station service and protection cable package

LV station service; CT/VT and breaker-control cable; trip/close DC; protection fiber; ground grid and equipment bonds.

Package focus: Define utility and GSU boundaries. HV underground cable is a separate engineered scope.

Specifies & buys: EPC / substation contractor via distribution; owner standards and utility team influence.

REFERENCES [59563399](#) [59563699](#) [59547599](#) [59945899](#)

03 Reel staging / prefab

Cable delivery and assembly work packages

All destination-zone cable families; circuit labels, reel/cut lists, bundles, kits and installation documentation.

Package focus: Offer preterminated assemblies only to an agreed engineered scope. Staging adds no circuit footage.

Specifies & buys: EPC construction, electrical contractor and distributor; assembly scope with the qualified packager.

REFERENCES Services and staging scope

02 Battery energy storage

Battery-to-collector interconnection package

RenewaFLEX for approved internal BESS wiring; MV collectors; PCS AC links; BMS/control, data and grounding.

Package focus: 13892403 is a 4/0, 2 kV BESS reference. 66131915 is a 35 kV EPR-insulated collector candidate only where the project design calls for it. Separate internal DC wiring from field AC scope.

Specifies & buys: BESS integrator owns internal design; EPC buys field AC, collector and site connections.

REFERENCES [13892403](#) [66131915](#)

04 Admin / control room

Control-room power and building systems

MV/LV supply; UPS interfaces; SIMpull THHN/THWN-2 branch wire; applicable Genesis building systems; data and grounding.

Package focus: Select fire alarm, security and network listings by system; qualify plant OS2 fiber separately.

Specifies & buys: EPC / electrical contractor for feeds; controls integrator and low-voltage contractor for systems.

REFERENCES [SIMpull Barrel THHN/THWN-2](#) [Genesis](#)

Click a stock reference for its technical document. Shared routes and staging do not add circuit footage. Illustrative model; cable design and quantities follow the approved project scope.

Applications 05–08. From equipment to scope.

05 E-house

Primary distribution and factory interface package

5 kV or 15 kV-class MV source/tie feeders; Royal Guard LV secondaries; control, protection, DC, data and bonding.

Package focus: MV examples: 55052299 (5/8 kV) and 55050899 (15 kV). Confirm the one-line and separate OEM internal wiring from EPC field connections.

Specifies & buys: E-house / switchgear OEM for internal wiring; EPC / distributor for field feeders and ties.

REFERENCES [55052299](#) [58950101 / 58950104](#) [55050899](#) [188040509](#)

07 Turbine hall

Turbine auxiliary and critical-power package

MV starting and pump feeds; LV lube-oil, turning-gear and vacuum loads; DLO emergency DC; control and temperature signals.

Package focus: Respect turbine OEM interfaces. Generator isolated-phase bus is separate.

Specifies & buys: Turbine OEM defines internal boundaries; EPC / electrical contractor buys external auxiliary circuits.

REFERENCES [167022R108 / 167024R108](#) [59212899](#)

06 HRSG trains & stacks

Repeat train power and instrumentation package

MV pump and transformer feeds; LV motors; VFD outputs where fitted; control, shielded pairs, RTD and thermocouple extension.

Package focus: Package by train. EPR/CPE reference 59548199 contains four #16 triads; select the required sensor circuit count. Qualify hot-zone leads and heat tracing separately.

Specifies & buys: HRSG / auxiliary package OEM for internal scope; EPC electrical and I&C teams for field wiring.

REFERENCES [59212899](#) [56295201](#) [59548199](#)

08 GT air inlets

Repeat inlet-house motor and sensor package

LV fan and pump feeders; drive output where fitted; damper control, differential-pressure signals, RTD and heating feeds.

Package focus: Confirm the modeled 3/C #2 feeder and motor count with the inlet OEM.

Specifies & buys: Turbine / inlet-house OEM and its packager; EPC for external supply connections.

REFERENCES Services and staging scope

Click a stock reference for its technical document. Shared routes and staging do not add circuit footage. Illustrative model; cable design and quantities follow the approved project scope.

Applications 09–12. From equipment to scope.

09 MCC / VFD / UPS

Motor-control and essential-power package

LV distribution; shielded VFD motor cable and terminations; UPS input/output and bypass; DLO battery feeds; control and bonding.

Package focus: Match output cable to the drive. Identify redundant AC and DC circuits.

Specifies & buys: MCC / VFD / UPS OEM for internal assembly; EPC and electrical contractor for field terminations.

REFERENCES [64346799](#) [167022R108 / 167024R108](#) [59193199](#) [188040509](#)

11 Air-cooled condenser

Repeat fan-bank power and controls package

MV transformer feeds; 500 kcmil LV secondaries; 3/C 250 kcmil fan-bank feeders; control, temperature signals and bonding.

Package focus: Use 59199899 for the fixed feeder candidate; select VFD cable where the fan is drive-fed.

Specifies & buys: ACC OEM / fan-drive suppliers for package wiring; EPC for transformer and field motor feeds.

REFERENCES [58950101 / 58950104](#) [59199899](#) [89052099](#)

10 Modular power yard

Repeat genset, black-start and fuel-cell interfaces

MV collectors; LV emergency and auxiliary power; OEM-approved starting-battery or fuel-cell DC; synchronization, control and data.

Package focus: Build repeat genset / RICE / fuel-cell interface kits from released drawings.

Specifies & buys: Genset / RICE / fuel-cell OEM and integrator for internal wiring; EPC for yard interconnects.

REFERENCES [167022R108 / 167024R108](#)

12 Chillers / cooling towers

Cooling-system motor and control package

MV or LV pump and compressor feeders; fan/pump drive outputs; control, RTD, heater feeds and equipment bonds.

Package focus: Reconcile the cooling concept with the final motor list before multiplying packages.

Specifies & buys: Chiller / cooling tower OEM, mechanical packager and EPC electrical contractor.

REFERENCES [64346799](#)

Click a stock reference for its technical document. Shared routes and staging do not add circuit footage. Illustrative model; cable design and quantities follow the approved project scope.

Applications 13–16. From equipment to scope.

13 Water / wastewater

Pump, treatment skid and process-control package

MV pumps; LV RO, demineralization, chemical and effluent systems; drive output; level, flow, pressure and analyzer signals.

Package focus: Check chemical exposure and submersion. Dedicated pump leads may be needed.

Specifies & buys: Water treatment OEM / process skids; EPC electrical and I&C contractor for interconnects.

REFERENCES [59212899](#)

15 Gas metering

Fuel-gas supply power and protection package

MV compressor supply; LV heaters and actuators; metering and shutdown signals; thermocouple/RTD; data and bonding.

Package focus: Consider ARMOR-X MC-HL where required. Match cable, glands and seals to area classification.

Specifies & buys: Fuel-gas skid / compressor supplier and EPC; process safety and electrical engineers set requirements.

REFERENCES [89057289](#) / [89057399](#) [59563699](#) [89066999](#)

14 Carbon capture

Capture auxiliary power and instrumentation package

MV compressor/blower feeds; LV pumps; VFD outputs; control, shielded instrumentation, analyzers, data and bonding.

Package focus: Develop a separate CCS schedule. 15 kV ARMOR-X candidates include 89066999 and three-ground 45834799. Confirm motor voltage, drive requirements and the complete hazardous-location installation.

Specifies & buys: CCS technology provider / compressor and pump OEMs; EPC for field wiring and plant ties.

REFERENCES [89057289](#) / [89057399](#) [89066999](#) [45834799](#)

16 Shared cable corridor

Plant-wide feeder and routing work packages

MV, LV, control, instrumentation, essential power, fiber and grounding routes connecting the plant zones.

Package focus: Create pull sections and reel plans. Count each circuit once. Select 5/8, 15 or 35 kV constructions from the actual system voltage and routing design.

Specifies & buys: EPC electrical / routing engineer, construction team and distributor; owner for resilience criteria.

REFERENCES [55052299](#) [55050899](#) [89008499](#) [67747299](#)

Click a stock reference for its technical document. Shared routes and staging do not add circuit footage. Illustrative model; cable design and quantities follow the approved project scope.

B / CABLE CATALOG

Medium-voltage power

Published references connect circuit requirements to specific constructions. Stock numbers do not indicate inventory.

STOCK # / REFERENCE	FAMILY	DUTY	CONSTRUCTION / SIZE	SPEC	DOCUMENTS	ZONES
55052299	■ MV power	5/8 kV MV-105	750 kcmil Cu; NL-EPR, tape shield, CPE-TP	46103	Spec 46103 · PDF Product page	05, 16
55050899	■ MV power	15 kV MV-105	500 kcmil Cu; NL-EPR 220 mil, 133%, tape shield, CPE-TP	46304	Spec 46304 · PDF Product page	05, 16
89008499	■ MV power	35 kV MV-105	750 kcmil Cu; NL-EPR 420 mil, 133%, tape shield, SIMpull PVC	46702	Spec 46702 · PDF Product page	16
67747299	■ MV power	35 kV MV-105 aluminum	1250 kcmil AL compact; NL-EPR 420 mil, 133%, tape shield, SIMpull PVC	46710	Spec 46710 · PDF Product page	16
66131915 Collector candidate only where the project design calls for it. Product page, no spec PDF published in the source.	■ MV power	35 kV EPR concentric-neutral primary	1/C 1000 kcmil aluminum; EPR 420 mil, 133%; copper concentric neutral; polyethylene jacket	35 kV EPR	Spec 35 kV EPR · no PDF in source Product page	02
89066999	■ MV power MC-HL · hazardous location	ARMOR-X MC-HL 15 kV	3/C 500 kcmil Cu; 15 kV, 133%; NL-EPR; tape shields; 1 × #1 ground; ARMOR-X / red PVC	46454	Spec 46454 · PDF Product page	14, 15

Reference types: product family (selected by size and listing) · specification number (a construction) · published base code (not a complete orderable stock number) · stock number (orderable; not inventory) · project-specific engineered selection. Type TC and TC-ER are distinct ratings.

Low-voltage, VFD and flexible / DC

Published references connect circuit requirements to specific constructions. Stock numbers do not indicate inventory.

STOCK # / REFERENCE	FAMILY	DUTY	CONSTRUCTION / SIZE	SPEC	DOCUMENTS	ZONES
45834799	■ VFD MC-HL · hazardous location	ARMOR-X MC-HL 15 kV VFD	3/C 350 kcmil Cu; 15 kV, 133%; NL-EPR; 3 × #6 grounds; ARMOR-X / red PVC; qualify drive system	46456	Spec 46456 · PDF Product page	14
58950101 / 58950104	■ LV power	Royal Guard 600 V	500 kcmil; EPR / thermoset CPE; 1,000 / 2,000 ft	45005	Spec 45005 · PDF Product page	05, 11
58949999	■ LV power	Royal Guard 600 V	350 kcmil; LV distribution single	45005	Spec 45005 · PDF	—
59199899	■ LV power	TC-ER motor feeder	3/C 250 kcmil + #4 ground; EPR XHHW-2 / CPE-TP	45255	Spec 45255 · PDF Product page	11
64346799	■ VFD	Machine Flex VFD	4 AWG representative; shielded XLPE / TPE	45440	Spec 45440 · PDF	09, 12
89052099	■ VFD MC-HL · hazardous location	ARMOR-X VFD	3/C 250 kcmil + 3 × #8 grounds; XLPE / PVC; MC-HL	45351	Spec 45351 · PDF Product page	11
167022R108 / 167024R108	■ Flexible / DC	DLO flexible power	1/C 2/0 or 4/0; 2 kV; EPDM / thermoset CPE	44020	Spec 44020 · PDF Product page	07, 09, 10
13892403 Approved internal BESS wiring. Separate internal DC wiring from field AC scope; not a solar PV application.	■ Flexible / DC	RenewaFLEX BESS	4/0 Cu; RHH/RHW-2, 2 kV; orange; 500 ft	44080	Spec 44080 · PDF Product page	02

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B / CABLE CATALOG

Control, instrumentation and protection

Published references connect circuit requirements to specific constructions. Stock numbers do not indicate inventory.

STOCK # / REFERENCE	FAMILY	DUTY	CONSTRUCTION / SIZE	SPEC	DOCUMENTS	ZONES
59563399	■ Control & I	CT/VT control	4/C #10; tinned Cu; EPR XHHW-2 / CPE	45059	Spec 45059 · PDF Product page	01
59563699	■ Control & I	CT/VT control	4/C #12; tinned Cu; EPR XHHW-2 / CPE	45059	Spec 45059 · PDF Product page	01, 15
59547599	■ Control & I	Breaker control	19/C #14; EPR / CPE	45059	Spec 45059 · PDF Product page	01
59193199	■ Control & I	Output / control	2/C #16; unshielded EPR XHHW-2 / CPE; Type TC	45059	Spec 45059 · PDF Product page	09
577883 (base code) Published base code, not a complete orderable stock number.	■ Control & I	Shielded output / control	2/C #16; foil shield; PVC/nylon / PVC	45154	Spec 45154 · PDF	—
59212899	■ Control & I	Analog / status pair	1 pair #18; overall shield; EPR / CPE	43618	Spec 43618 · PDF	06, 07, 13
56295201	■ Control & I	RTD triad candidate	1 triad #18; overall shield; PVC/nylon / PVC	43606	Spec 43606 · PDF	06
89057289 / 89057399	■ Control & I MC-HL · hazardous location	ARMOR-X instrumentation	2 / 4 pairs #18; armored MC-HL construction	43702	Spec 43702 · PDF	14, 15
59945899	■ Control & I	Substation control	4/C #14 tinned Cu; EPR XHHW-2 / CPE-TP; TC-ER; black/white/red/green	45059	Spec 45059 · PDF Product page	01
59548199 Four #16 triads. Match sensor count and jacket to the specification.	■ Control & I	EPR/CPE instrumentation triads	4 triads, #16 tinned Cu; individual + overall shields; EPR / CPE; TC-ER	43624	Spec 43624 · PDF Product page	06

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Type TC and TC-ER are distinct ratings.

Grounding, networks, building systems and specialty

Published references connect circuit requirements to specific constructions. Stock numbers do not indicate inventory.

STOCK # / REFERENCE	FAMILY	DUTY	CONSTRUCTION / SIZE	SPEC	DOCUMENTS	ZONES
188040509	Specialty	SIS switchboard wire	4 AWG 133-strand tinned Cu; XLPE; 600 V, 90 °C; switchgear and panel internal wiring	LW93	Spec LW93 · PDF Product page	05, 09
10704542 / 16266909	Grounding	Bare copper ground	4/0 7-strand and 2/0 37-strand published examples; final sizes follow the grounding study	10080	Spec 10080 · PDF	01, 16
SIMpull Barrel THHN/THWN-2 Family reference; select the stock number by size.	Specialty	Building branch wire	THHN/THWN-2 branch wire for building systems	10026	Spec 10026 · PDF	04
Genesis Product family page; no single stock number applies.	Specialty	Fire alarm, security and building-control cable	Select the actual system listing	—	Product page	04
VFD spec sheets 85451 / 85452 / 85456	Specialty	Terminations and accessories	MV terminations, lugs, glands, VFD shield connections, identification and supports	85451 / 85452 / 85456	Spec 85451 / 85452 / 85456 · PDF Spec 85452 · PDF Spec 85456 · PDF	—

Reference types: product family (selected by size and listing) · specification number (a construction) · published base code (not a complete orderable stock number) · stock number (orderable; not inventory) · project-specific engineered selection. Type TC and TC-ER are distinct ratings.

C / DOCUMENTED EXPERIENCE · RESOURCE LIBRARY

Additional cases with a reference.

Historical Southwire examples. The PowerGen application is a proposed use, not an outcome from the illustrated plant.

ONSHORE HV CONNECTION / VINEYARD WIND 1 32 miles High-voltage cable, manufactured and installed. CUSTOMER PROBLEM An onshore high-voltage cable system for a generation project's grid connection. SOUTHWIRE CONTRIBUTION Design, manufacturing and installation of the project's onshore high-voltage cable system. PUBLISHED RESULT Southwire's case study describes design, manufacturing and installation of the project's onshore high-voltage cable system. POWERGEN APPLICATION · PROPOSED Engage the HV team early when a generation project requires a qualified underground gen-tie or grid connection. Read the case study · PDF	URD RELIABILITY PROGRAM / NV ENERGY \$800 million Replacement cost deferred. CUSTOMER PROBLEM A large population of aging underground distribution cable. SOUTHWIRE CONTRIBUTION A multi-year rejuvenation program described in a Southwire-hosted paper. PUBLISHED RESULT The paper reports over 12 million conductor feet rejuvenated through 2020 and the utility's estimate of more than \$800 million in deferred replacement cost. POWERGEN APPLICATION · PROPOSED Use condition assessment to compare rejuvenation, repair and replacement for eligible aging plant feeders. Utility program results are not a forecast for a generation site. Read the case study · PDF	69 KV TRANSMISSION / ATLANTIC MUNICIPAL UTILITIES 1969 vintage A 69 kV underground circuit kept in service. CUSTOMER PROBLEM A 1,300-foot underground 69 kV section originally installed in 1969. SOUTHWIRE CONTRIBUTION Rejuvenation of the existing 500 kcmil aluminum, 69 kV XLPE cable. PUBLISHED RESULT Southwire's paper describes rejuvenation of the section rather than replacement. POWERGEN APPLICATION · PROPOSED For suitable brownfield circuits, evaluate cable condition, accessories and outage requirements before choosing rejuvenation or replacement. Read the case study · PDF
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